4.6.2 The inverse cosecant function

The cosecant function, $\operatorname{cosec} : \left[-\frac{\pi}{2}, 0\right) \cup \left(0, \frac{\pi}{2}\right] \to (-\infty, -1] \cup [1, \infty)$ is bijective in the restricted domain $\left[-\frac{\pi}{2}, 0\right) \cup \left(0, \frac{\pi}{2}\right]$. So, the inverse cosecant function is defined with the domain $(-\infty, -1] \cup [1, \infty)$ and the range $\left[-\frac{\pi}{2}, 0\right) \cup \left(0, \frac{\pi}{2}\right]$.

Definition 4.6

The inverse cosecant function $\operatorname{cosec}^{-1} : (-\infty, -1] \cup [1, \infty) \to \left[-\frac{\pi}{2}, 0\right) \cup \left(0, \frac{\pi}{2}\right]$ is defined by $\operatorname{cosec}^{-1}(x) = y$ if and only if $\operatorname{cosec} y = x$ and $y \in \left[-\frac{\pi}{2}, 0\right) \cup \left(0, \frac{\pi}{2}\right]$.

4.6.3 Graph of the inverse cosecant function

The inverse cosecant function, $y = \operatorname{cosec}^{-1} x$ is a function whose domain is $\mathbb{R} \setminus (-1, 1)$ and the range is $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \setminus \{0\}$. That is, $\operatorname{cosec}^{-1} : (-\infty, -1] \cup [1, \infty) \to \left[-\frac{\pi}{2}, 0\right) \cup \left(0, \frac{\pi}{2}\right]$.

Fig. 4.21 and Fig. 4.22 show the graphs of cosecant function in the principal domain and the inverse cosecant function in the corresponding domain respectively.

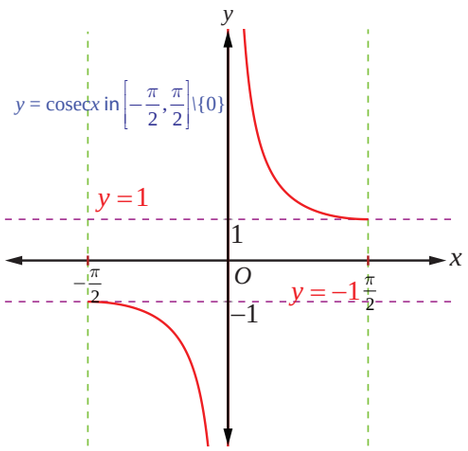


Fig. 4.21

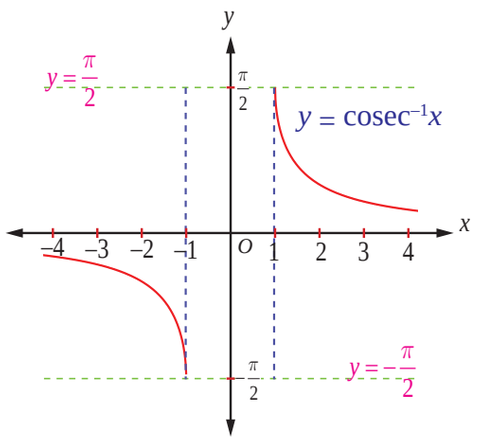


Fig. 4.22

4.7 The Secant Function and Inverse Secant Function

The secant function is defined as the reciprocal of cosine function. So, $y = \sec x = \frac{1}{\cos x}$ is defined for all values of x except when $\cos x = 0$ .Thus, the **domain** of the function $y = \sec x$ is $\mathbb{R} \setminus \left\{(2n+1)\frac{\pi}{2} : n \in \mathbb{Z}\right\}$. As $-1 \le \cos x \le 1$, $y = \sec x$ does not take values in $(-1, 1)$. Thus, the range of the secant function is $(-\infty, 1] \cup [1, \infty)$. The secant function has neither maximum nor minimum. The function $y = \sec x$ is a periodic function with period 2π and it is also an even function.